Lab Report :Automated measurement of fetal head circumference using 2D ultrasound images

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I. Introduction

Fetal head circumference (HC) is an important biometric measurement used in prenatal care to monitor fetal growth and development. Traditionally, HC is measured manually by medical professionals using ultrasound images, which can be time-consuming and depends on operator experience. In this project, a deep learning–based regression model is developed to automatically estimate fetal head circumference from ultrasound images. To improve accuracy, both the ultrasound image and the pixel size information are used as inputs to the model. The goal of this project is to design, train, and evaluate a convolutional neural network (CNN) that can predict head circumference values in millimeters.

II. Dataset and Preprocessing

The dataset used in this project consists of fetal ultrasound images along with a CSV file containing metadata. Each image represents a single ultrasound scan of the fetal head. The CSV file includes the image filename, pixel size in millimeters, and the corresponding head circumference value in millimeters. The images are stored in separate folders for training and testing, with the training set containing approximately 1998 images. All images are grayscale and vary in size, so they are resized to a fixed resolution before being used in the model. The head circumference value serves as the regression target for training.

III. Methodology and Implementation

A. Data Preparation

The dataset contains fetal ultrasound images and a CSV file providing pixel size and head circumference values in millimeters. All images were loaded in grayscale format and resized to a fixed resolution of 224×224 pixels to ensure consistent input dimensions for the convolutional neural network. Pixel intensity values were normalized to the range $[0, 1]$ to improve numerical stability during training.

Pixel size values were extracted from the CSV file and normalized using standard scaling. This step ensures that the numerical pixel size input has a similar scale to the learned image features, which helps the model train more effectively. The dataset was then split into 80% training data and 20% validation data to evaluate model performance on unseen samples.

B. Multi-Input Learning Strategy

Head circumference estimation depends on both the visual appearance of the fetal head and the physical scale of the ultrasound image. To account for this, a multi-input learning strategy was implemented. One input branch processes the ultrasound image, while a second input branch processes the pixel size value. This design allows the model to learn anatomical features from the image and combine them with real-world measurement information.

The pixel size input is treated as a numerical feature and passed through fully connected layers to learn its contribution to the final head circumference prediction.

C. Model Architecture

I built a multi-input Convolutional Neural Network (CNN) to estimate fetal head circumference from ultrasound images. The model uses both image data and pixel size information to improve prediction accuracy. The architecture includes:

- **2D Convolutional Layers (Image Branch):** These layers extract important visual features from ultrasound images, such as edges and head contours. Multiple convolutional layers with increasing filters are used to learn both low-level and high-level features.
- **Batch Normalization and Max Pooling:** Batch normalization helps stabilize training, while max pooling reduces the spatial size of feature maps and allows the model to focus on the most important features.
- **Global Average Pooling:** This layer converts feature maps into a compact feature vector and reduces overfitting.
- **Pixel Size Branch:** A separate fully connected branch processes the pixel size value (mm per pixel), allowing the model to learn the effect of image scale on head circumference.
- **Feature Fusion and Output:** The image features and pixel size features are concatenated and passed through fully connected layers. The final output layer uses linear activation to predict the head circumference in millimeters.

D. Training and Results

The model was trained using the Adam optimizer and Mean Absolute Error (MAE) as the evaluation metric. The training settings are summarized as follows:

- **Epochs:** 40
- **Batch Size:** 16
- **Validation:** 20% of the dataset was used for validation

Data augmentation was applied during training to reduce overfitting. The initial image-only CNN achieved a validation MAE of approximately 44 mm. After introducing the multi-input architecture with pixel size information, the validation MAE improved to approximately 20.97 mm, demonstrating the effectiveness of the proposed approach.

IV. Results and Discussion

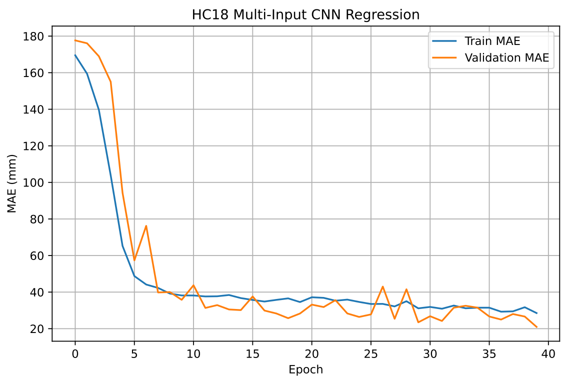

Fig. 1. Training and validation MAE curves for the HC18 multi-input CNN model.

Figure 1 shows the training and validation Mean Absolute Error (MAE) of the proposed multi-input CNN model over the training epochs. As training progresses, both the training MAE and validation MAE gradually decrease, indicating that the model is able to learn meaningful features from the ultrasound images and pixel size information.

The training MAE decreases steadily, showing that the model fits the training data well. The validation MAE follows a similar trend and stabilizes after several epochs, which suggests that the model generalizes reasonably well to unseen data and does not suffer from severe overfitting. Minor fluctuations in the validation curve are expected due to the limited size and variability of the dataset.

The final validation MAE achieved by the multi-input CNN is approximately 20.97 mm. Compared to the image-only CNN, which achieved a validation MAE of around 44 mm, the inclusion of pixel size information significantly improves prediction accuracy. This result demonstrates that pixel size plays an important role in converting image-based features into real-world head circumference measurements.

Although the error is still higher than clinical standards, the results show that the proposed approach is effective for automatic fetal head circumference estimation and suitable for a student-level implementation. Further improvements could be achieved by using more advanced architectures or incorporating segmentation-based methods.

V. Comparition

Based on the comparison with the HC18 Challenge leaderboard provided by Mr. Son, the proposed multi-input CNN model achieved a final Mean Absolute Error (MAE) of approximately 20.97 mm. This result places the model at around 1765th position on the public leaderboard. Although the ranking is not among the top-performing methods, the result is still reasonable and acceptable for a student-level implementation. However, the performance gap compared to leading solutions indicates that further improvements are possible, such as using more advanced architectures, better data augmentation.